

Exploring and Testing Social Science Theories by using Agent-Based Modeling and Simulations

Assignment 4

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Subject

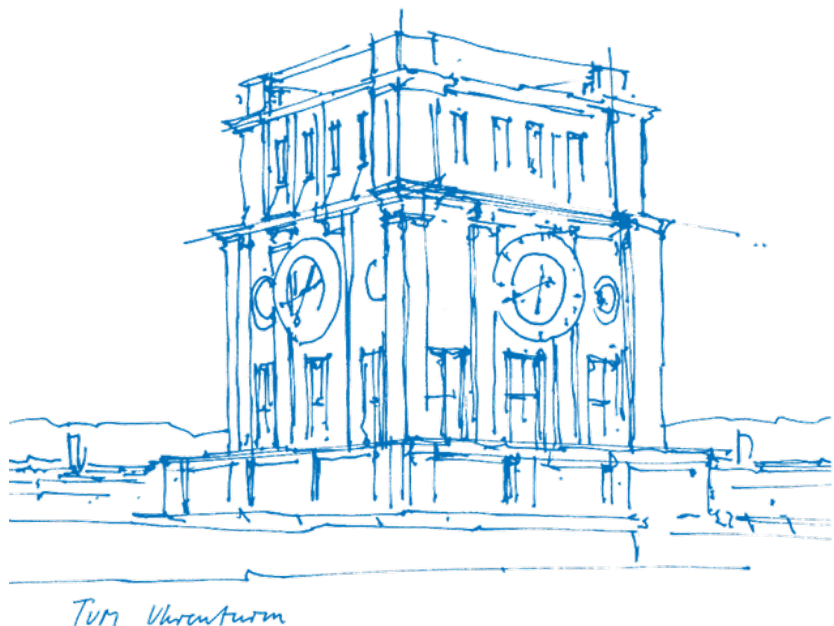
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Examiner

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1 Methodology & Reproducibility

1.1 Introduction

Agent-based models (ABMs) test social science theories by translating hypotheses into executable agent-level rules and observing emergent macro-level patterns (Gilbert & Troitzsch, 2005; Squazzoni et al., 2014). This report applies two NetLogo ABMs to investigate two theories. First, Hammond and Axelrod’s ethnocentrism model (R. . Hammond & Axelrod, 2006) operationalizes *homophily*, the tendency for similar individuals to associate (McPherson et al., 2001), by giving agents a binary cooperate/defect gene for same-type and different-type neighbors, and asks whether ethnocentric strategies dominate under evolutionary selection. Second, Epstein’s civil violence model (Epstein, 2002) represents rebellion as a threshold decision where agents weigh grievance against perceived risk of arrest, enabling controlled tests of whether movement affects rebellion frequency, how gradual versus sudden legitimacy collapse shapes civil unrest, and whether Tocqueville’s claim, that revolutions arise from relaxed repression rather than worsening conditions, can be reproduced (de Tocqueville, 1980).

Both models are deliberately simplified, omitting communication networks, organizational structure, and elite dynamics. Following Saam’s distinction between theoretical exploration and empirical prediction (Saam, 2017), we treat them as tools for the former, and compare outcomes to historical cases with appropriate caution (Larooij & Törnberg, 2026; Sargent, 2010).

1.2 Methodology

1.2.1 Software

Simulations were run in NetLogo 7.0.4 via BehaviorSpace (command-line batch execution). Output CSVs were analyzed in Python (NumPy, SciPy, Matplotlib, Pandas). All code, experiment definitions, and raw data are included in the submission.

1.2.2 Model 1: Ethnocentrism

Agents on a toroidal grid belong to one of four color groups and carry two binary strategy genes (cooperate-with-same, cooperate-with-different), producing four types: *Altruist* (CC), *Ethnocentric* (CD), *Cosmopolitan* (DC), and *Egoist* (DD). Cooperation transfers a fitness benefit (GAIN-OF-RECEIVING) at a cost (COST-OF-GIVING). Higher-fitness agents reproduce into adjacent patches; offspring inherit genes with a small mutation probability. Immigrants arrive each tick with strategy probabilities set by two sliders.

Table 1

Ethnocentrism experiments.

Exp.	Variation	Runs	Ticks
Q1 Default	All NetLogo defaults, setup empty	10	2,000
Q3 Altruist-only	Both cooperation sliders = 1.0	20	2,000
Q4 Cost/gain grid	Gain $\in \{.01, .03, .06\} \times$ Cost $\in \{0, .01, .03, .06\}$	5/cell	2,000

Metrics: population share of each strategy, cooperation rate, and ethnocentric-consistent interaction rate, recorded at tick checkpoints (100, 500, 1000, 1500, 2000).

1.2.3 Model 2: Rebellion

Citizens on a grid have fixed hardship $H_i \sim U(0, 1)$ and risk aversion $R_i \sim U(0, 1)$. Grievance is $G_i = H_i(1 - L)$ where L is government legitimacy. A citizen rebels when $G_i - R_i \cdot P_i \cdot J_{\max} > T$, where P_i is the local cop-to-rebel ratio and $J_{\max}$ the maximum jail term. Cops arrest one active rebel per tick. Default settings: 40×40 grid, 1,120 citizens

(70% of 1,600 patches), 64 cops (4%), vision 7, $J_{\max} = 30$, $L = 0.82$. A rebellion *episode* is any tick with ≥ 25 active agents.

Table 2

Rebellion experiments.

Exp.	Variation	Runs	Ticks	Purpose
Q5 Movement	MOVEMENT? on/off	20/cond.	1,000	Epstein's claim
Q6 Gradual	L : $0.9 \rightarrow 0$ over 250 ticks	10	250	Sustained erosion
Q6 Sudden	L : 0.9 for 80t, drop to 0.7	10	250	Shock dynamics
Q7 Legit. eros.	L : $0.82 \rightarrow 0$, cops fixed	10	250	Worsening conditions
Q7 Cop erosion	Cops: $64 \rightarrow 0$, L fixed	10	250	Relaxing repression

Per-tick metrics: active agents, jailed agents, cops, legitimacy. Per-run aggregates: episodes, mean/peak active, rebellion share, mean jailed, onset tick, final cops/legitimacy.

1.2.4 Statistical Analysis

For each paired comparison (Q5–Q7) we report **Welch's t -test** (unequal-variance, robust to heteroscedasticity), **Mann-Whitney U** (non-parametric robustness check for small n), and **Cohen's d** as the primary effect size measure:

$$d = \frac{\bar{x}_A - \bar{x}_B}{s_p}, \quad s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}} \quad (1.1)$$

interpreted as negligible ($|d| < 0.2$), small (0.2–0.5), medium (0.5–0.8), or large (> 0.8). Significance threshold: $\alpha = 0.05$. A cross-question sensitivity analysis compares the largest $|d|$ across Q5–Q7 to identify which parameters the model is most sensitive to.

1.2.5 Limitations

At $n = 10$ per condition (Q6–Q7), t -tests have moderate power for large effects; Cohen's d remains interpretable regardless of sample size and is the primary basis for conclusions. Extremely large d values (> 3) in Q6–Q7 partly reflect low within-group variance from deterministic parameter schedules rather than massive between-group differences. Both models omit communication networks, organizational structure, media, elite dynamics, and external actors; results demonstrate that tested mechanisms are *theoretically sufficient*, not that they are the sole real-world drivers.

1.3 Code

The complete source code developed for this assignment, including all NetLogo models, simulation experiments, data analysis scripts, and generated results, has been made publicly available to ensure transparency and reproducibility. The repository can be accessed on Github [LINK](#).

2 Assignment Questions

2.1 Exploring the Theory of Homophily

Question 1

[25 points]

Using the default settings (including `setup empty`), analyze the evolution of each of the four strategies: cooperate with everyone, cooperate only with people of the same type, cooperate with no one, and cooperate only with people of different types. Run the simulation multiple times to obtain robust findings. Describe and, if possible, explain your results. Interpret your findings with respect to the theory of homophily.

Results

Using the default settings (including `setup empty`), the simulation was run 10 times to obtain robust results. The outcomes were highly consistent across runs. At tick 2000, the average population shares were:

Table 3

Q1: Final strategy shares under default settings (10 runs, tick 2000).

Strategy	Behavior	Avg. Share
Altruists (CC)	Cooperate with everyone	15.2%
Ethnocentrics (CD)	Cooperate only with same type	75.0%
Cosmopolitans (DC)	Cooperate only with different types	2.2%
Egoists (DD)	Cooperate with no one	7.6%

Altruists (CC) were relatively common at the beginning of the simulation. Their proportion was about 40.1% at tick 100 but steadily declined over time before stabilizing

around 15.2%. Although altruists maximize cooperation, they also pay the cost of helping every individual regardless of group membership. This makes them vulnerable to exploitation by less cooperative strategies, reducing their evolutionary success. (Gross et al., 2026)

Ethnocentrics (CD) agents showed the opposite trend. They increased from 26.6% at tick 100 to approximately 75.0% by tick 2000, becoming the dominant strategy in every simulation. Ethnocentrics cooperate only with members of their own group. As agents become spatially clustered through reproduction, they receive the benefits of cooperation within their own group while avoiding the costs of helping outsiders. This provides the highest overall fitness under the default parameters.

Cosmopolitans (DC) agents declined rapidly from 17.2% at tick 100 to only 2.2% at the end of the simulation. Because they cooperate only with members of different groups, they rarely benefit from reciprocal cooperation once homophilic clusters emerge. They continue paying cooperation costs without receiving comparable benefits, making this strategy evolutionarily unstable. (Gross et al., 2023)

Egoists (DD) decreased moderately from 16.2% to around 7.6%. They avoid all cooperation costs but also receive none of the benefits generated by cooperative interactions. Since cooperation is beneficial under the default benefit-cost ratio, egoists perform worse than ethnocentrics, although they survive as a small minority.

Interpretation with Respect to Homophily

The simulation strongly supports the theory of **homophily**. Although the model does not explicitly force segregation, local reproduction and interactions naturally create clusters of similar agents. Within these clusters, ethnocentric agents cooperate frequently with one another, generating mutual benefits while avoiding costly interactions with outsiders.

This combination of high internal cooperation and limited external cooperation gives ethnocentric agents a consistent evolutionary advantage over both universal cooperators (altruists) and non-cooperators (egoists). (Hartshorn et al., 2013)

The decline of cosmopolitan agents further illustrates this mechanism. Their strategy depends on interactions with different groups, but increasing homophily reduces such opportunities, causing their fitness to decline.

The repeated simulation runs demonstrate that even without explicit hostility toward other groups, a simple preference for cooperating with similar individuals can generate strong in-group favoritism. Small local interaction preferences therefore scale into persistent group clustering and the dominance of ethnocentric behavior, providing a computational illustration of homophily theory.

Question 2

[15 points]

Relate your findings from Question 1 to the field of Science of Science (discussed during the first three weeks of class). What do your findings imply for: an individual researcher when selecting co-authors, funding agencies, and the progress of science over time? Explain your answers.

Results

The findings from Question 1 suggest that homophily influences not only social interactions but also the organization of scientific collaboration. Researchers tend to collaborate with others who share similar backgrounds, expertise, institutions, or research interests.

Individual Researchers selecting co-authors with similar expertise can lower coordination costs because collaborators often share a common language, methodological approaches, and research culture. This facilitates communication and increases produc-

tivity. However, relying exclusively on similar collaborators may reduce exposure to novel perspectives and interdisciplinary knowledge, potentially limiting creativity and breakthrough discoveries.

Funding Agencies The model suggests that funding agencies should encourage collaborations across disciplinary and institutional boundaries. Borsato and Erasmo (2026) argue that funding agencies should balance research excellence with diversity to foster innovation. This can be achieved through interdisciplinary grant programs, international mobility schemes, and funding incentives for diverse research teams.

Scientific Progress Homophily contributes to the formation of specialized research communities, which can accelerate progress within individual fields by fostering trust and efficient collaboration. Specht and Crowston (2022) show that interdisciplinary collaborations between diverse research teams are associated with significant scientific outcomes. If collaboration becomes too concentrated within homogeneous groups, science risks becoming fragmented. Limited interaction between communities may lead to duplicated research efforts, slower diffusion of knowledge, and fewer interdisciplinary innovations. Scientific progress benefits from balancing strong within-field collaboration with opportunities for cross-disciplinary exchange.

Question 3

[5 points]

Set both IMMIGRANT-CHANCE-COOPERATE sliders to 1 . 0. This means that only altruists are initially created. Do ethnocentrism and the other two strategies ever evolve? If so, do they ever out-compete the altruists?

Results

To investigate whether ethnocentrism can emerge from a purely altruistic population, both IMMIGRANT-CHANCE-COOPERATE sliders were set to 1 . 0. Under this setting, all newly created immigrants initially followed the altruistic strategy.

Despite these initial conditions, ethnocentric, cosmopolitan, and egoist strategies still evolved through mutation. Across 20 simulation runs, ethnocentric agents became the dominant strategy, accounting for an average of 68.4% of the population at tick 2000, while altruists represented only 24.6%. Ethnocentric agents outnumbered altruists in all 20 simulation runs.

Table 4

Q3: Final-state averages for the altruist-only immigrant condition (20 runs, tick 2000).

Metric	Mean	SD	Min	Max
Population size	1645.2	58.92	1560	1756
CC (Altruists)	24.6%	5.6%	17.6%	43.8%
CD (Ethnocentrics)	68.4%	6.3%	48.2%	76.2%
DC (Cosmopolitans)	1.7%	1.0%	0.4%	3.8%
DD (Egoists)	5.3%	1.1%	2.9%	7.1%
Ethnocentric interactions	88.2%	2.2%	83.2%	91.3%
Overall cooperation	80.5%	1.8%	77.4%	85.3%

These findings demonstrate that a fully altruistic initial population is not evolutionarily stable. Mutation continually introduces alternative strategies, and natural selection favors ethnocentric agents because they retain the benefits of cooperation within their own group while avoiding the costs of cooperating with outsiders. (Hartshorn et al., 2013) ethnocentrism consistently outcompetes universal cooperation, even when every immigrant initially enters the population as an altruist.

Question 4**[5 points]**

Change the values of COST-OF-GIVING and GAIN-OF-RECEIVING, and observe the effects on the model and the level of ethnocentricity. Interpret your findings.

Results**Table 5**

Q4: Average final-state outcomes for different combinations of cooperation gain and cost (5 runs per setting).

Gain	Cost	CC	CD	DC	DD	Ethno.	Coop.	Pop.
0.01	0.00	0.401	0.482	0.052	0.065	0.817	0.811	1340.8
0.01	0.01	0.159	0.220	0.227	0.394	0.398	0.372	656.4
0.01	0.03	0.036	0.052	0.267	0.645	0.133	0.093	554.8
0.01	0.06	0.015	0.029	0.170	0.787	0.095	0.038	429.0
0.03	0.00	0.207	0.739	0.017	0.036	0.910	0.825	1820.4
0.03	0.01	0.161	0.727	0.024	0.088	0.878	0.764	1629.2
0.03	0.03	0.087	0.193	0.181	0.539	0.326	0.275	610.4
0.03	0.06	0.018	0.035	0.155	0.791	0.119	0.049	516.0
0.06	0.00	0.209	0.746	0.012	0.033	0.932	0.861	2109.4
0.06	0.01	0.166	0.774	0.013	0.046	0.920	0.823	2080.8
0.06	0.03	0.059	0.786	0.018	0.137	0.865	0.715	1761.6
0.06	0.06	0.048	0.152	0.087	0.713	0.266	0.176	690.8

Note: CC = Altruists, CD = Ethnocentrics, DC = Cosmopolitans, DD = Egoists. Ethno. = proportion of ethnocentric-consistent interactions. Coop. = overall cooperation rate. Pop. = final population size.

When the benefits of cooperation exceeded the costs, ethnocentric agents consistently dominated the population. For example, with a gain of 0.06 and a cost of 0.03, ethnocentric agents accounted for 78.6% of the population while the overall cooperation rate remained relatively high (71.5%). Restricting cooperation to members of the same group allowed obtaining the benefits of cooperation while minimizing unnecessary costs.

In contrast, when cooperation costs became as high as the benefits (gain = 0.06, cost = 0.06), cooperation largely collapsed. Egoists became the dominant strategy (71.3%), the overall cooperation rate dropped to only 17.6%, and the population size declined substantially. Cooperation survives when rewards exceed costs, but indiscriminate cooperation is vulnerable. Ethnocentrism is favored under many benefit-cost settings (R. A. Hammond & Axelrod, 2006) because it preserves in-group cooperation while reducing costly out-group giving.

2.2 Exploring Dynamics of Civic Unrest

Question 5

[5 points]

Epstein claims that movement of the agents causes more frequent episodes of rebellion. Run the model with the `MOVEMENT?` switch turned on and off. Can you verify Epstein's claim? Explain your findings.

Results

We tested Epstein (2002)'s claim by running the Rebellion model with the `MOVEMENT?` switch toggled on and off, holding all other parameters at their defaults (`GOVERNMENT-LEGITIMACY` = 0.82, initial cop density = 4%, initial citizen density = 70%). Twenty independent replications were conducted per condition, each running for 1,000 ticks.

Table 6

Q5: Movement on vs. off aggregate comparison across 20 runs per condition.

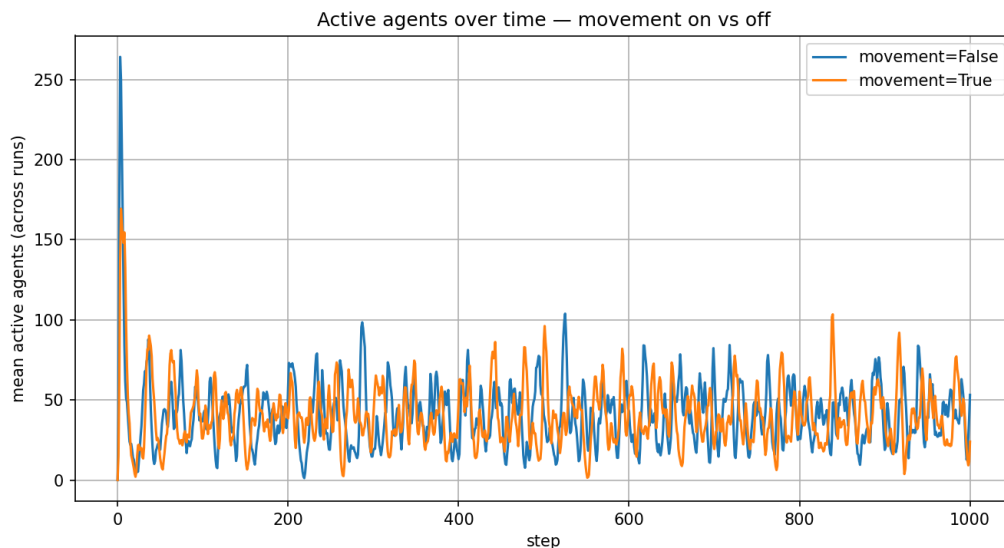
Metric	Move ON	SD	Move OFF	SD	Cohen's <i>d</i>	<i>p</i> (Welch)
Mean episodes	37.55	2.76	37.75	4.01	−0.06	0.855
Mean active agents	40.32	4.38	41.18	3.81	−0.21	0.514
Peak active agents	348.6	20.30	353.6	17.61	−0.26	0.411
Share in rebellion	0.267	0.02	0.275	0.02	−0.42	0.191
Mean jailed agents	234.2	15.96	241.2	15.70	−0.44	0.173

No comparison reaches statistical significance ($p < 0.05$), and all effect sizes are negligible to small ($|d| < 0.5$). The time-series traces of active agents are nearly

indistinguishable (Figure 1). Mann–Whitney U tests confirm the parametric results (all $p > 0.11$).

Figure 1

Active agents over time with movement on (orange) vs. off (blue). The two traces overlap almost entirely, indicating no substantive effect of movement.



We therefore **cannot verify Epstein’s claim** under these conditions. One plausible explanation is that at the default density settings (70% citizens, 4% cops), the grid is already densely populated, so spatial mixing through movement provides little additional contact between agents and cops beyond what already occurs through adjacency. Movement may matter more at lower densities, where spatial clustering could otherwise shield rebels from police detection. This is consistent with Epstein (2002)’s original observation being sensitive to parameterization, and it highlights the importance of sensitivity analysis in simulation-based research (Sargent, 2010).

Question 6

[5 points]

Compare the effects of a gradual erosion in the perceived legitimacy of the central authority to a smaller, but immediate drop. For example, set GOVERNMENT-LEGITIMACY

to 0.9 and reduce it to 0 over 250 runs, plotting the legitimacy and the number of active agents. Now do another run, beginning with legitimacy at 0.9. This time, leave the legitimacy constant for 80 runs and then drop it to 0.7 at once.

- Describe your results.
- Try to explain your results.
- What are the implications of this result for the strategy of revolutionary leaders?

Results

We compared two legitimacy-change scenarios, each run for 250 ticks with 10 replications. In the *gradual erosion* condition, government legitimacy was reduced linearly from 0.9 to 0 over 250 ticks. In the *sudden drop* condition, legitimacy was held constant at 0.9 for 80 ticks, then dropped instantaneously to 0.7.

Table 7

Q6: Gradual erosion vs. sudden legitimacy drop aggregate comparison (10 runs each).

Metric	Gradual	SD	Sudden	SD	Cohen's <i>d</i>	<i>p</i> (Welch)
Mean episodes	8.70	1.49	8.10	1.20	+0.44	0.336
Mean active agents	107.0	10.59	72.7	6.58	+3.90	<0.001
Peak active agents	432.2	46.09	631.5	12.74	−5.89	<0.001
Share in rebellion	0.688	0.03	0.360	0.02	+12.45	<0.001
Mean jailed agents	586.2	12.99	315.5	11.35	+22.19	<0.001

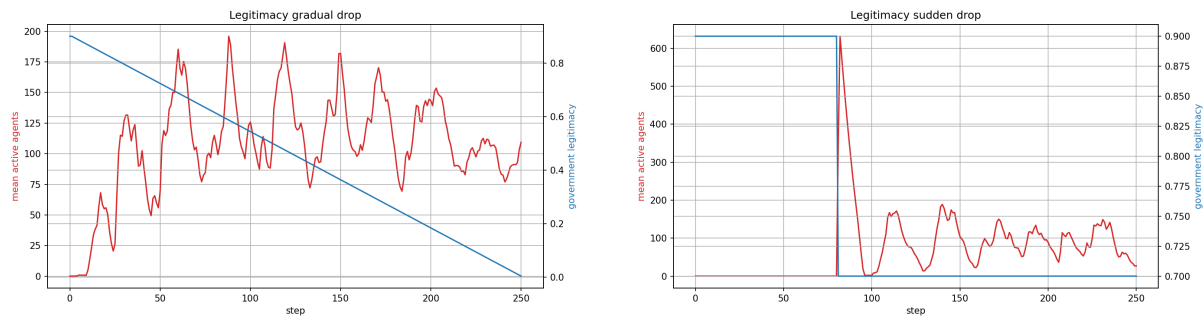
The two conditions produce highly significant differences across all metrics except episode count. The large Cohen's *d* values (all > 3) reflect both genuine between-condition differences and the low within-group variance that results from deterministic

parameter schedules. Mann–Whitney U tests confirm all significant results ($p < 0.001$).

Figure 2 shows the contrasting dynamics.

Figure 2

Q6: Active agents (red) and government legitimacy (green/orange) over time. Left: gradual erosion from 0.9 to 0. Right: sudden drop from 0.9 to 0.7 at tick 80.



Description of results. **Gradual erosion** produces sustained, escalating unrest. As legitimacy declines linearly, the baseline number of active agents rises progressively, and rebellion episodes increase in amplitude over time. Agents spent 68.8% (SD = 3%) of the simulation in active rebellion, with a mean of 107.0 active agents (SD = 10.6) and a peak of 432.2 (SD = 46.1).

Sudden drop produces a single dramatic spike. When legitimacy drops from 0.9 to 0.7 at tick 80, there is an immediate and massive surge in active agents the peak reaches 631.5 (SD = 12.7), far exceeding the gradual condition's peak. However, this spike is transient: the system settles into a new equilibrium at the lower legitimacy level. Overall, agents spent only 36.0% (SD = 2%) of the time in rebellion, with a mean of 72.7 active agents (SD = 6.6).

Explanation. In the Epstein model, each citizen rebels when their grievance $G_i = H_i(1 - L)$ exceeds a risk-adjusted threshold $R_i \cdot P_i \cdot J_{\max}$. Under gradual erosion, the monotonic decrease in L continuously pushes more agents past their activation

threshold, producing an escalating but moderate pattern. Under sudden drop, many agents simultaneously cross the threshold at tick 80, producing a large but brief spike. Once arrests restore the local cop-to-rebel ratio, the system re-equilibrates (Epstein, 2002).

Implications for revolutionary strategy. These findings carry distinct strategic implications. A sudden shock to regime legitimacy such as a scandal, a massacre, or an economic crisis can mobilize massive numbers instantaneously, creating a narrow window of opportunity for regime change. However, if the regime survives the initial spike, the population adapts and rebellion subsides. Gradual erosion, by contrast, creates persistent and deepening instability that eventually becomes self-sustaining, suggesting that slow delegitimization (through sustained campaigning, media exposure, or institutional decay) may be more effective for sustained revolution, even though it lacks the dramatic spark of a sudden crisis. Revolutionary leaders therefore face a trade-off between *intensity* (sudden shocks, peak $d = -5.89$) and *duration* (gradual undermining, rebellion-share $d = +12.45$).

Question 7

[30 points]

Conduct a series of controlled experiments to test the following quote from Tocqueville, which Epstein cites:

“It is not always when things are going from bad to worse that revolutions break out. On the contrary, it oftener happens that when a people that has put up with oppressive rule over a long period without protest suddenly finds the government relaxing its pressure, it takes up arms against it.”

Test de Tocqueville’s claim by comparing the effects of gradual erosion in perceived legitimacy with gradual erosion in the number of cops. Describe and interpret your results. Can you verify Tocqueville’s claim?

Results

de Tocqueville (1980) argued that revolutions are more likely when an oppressive regime *relaxes its pressure* than when conditions are steadily worsening. Epstein (2002) cites this observation as a key feature his model should reproduce. We tested this by comparing two mechanisms of “relaxation,” each run over 250 ticks with 10 replications:

- **Legitimacy erosion** (worsening conditions): Government legitimacy was reduced linearly from 0.82 to 0 over 250 ticks, with the number of cops held constant at 64.
- **Cop erosion** (relaxing repression): The number of cops was reduced linearly from 64 to 0 over 250 ticks, with legitimacy held constant at 0.82.

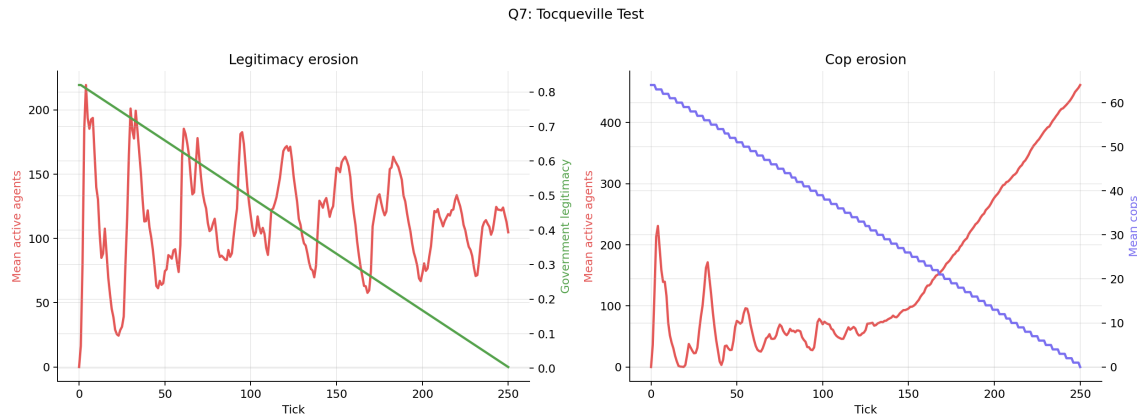
Table 8

Q7: Legitimacy erosion vs. cop erosion aggregate comparison (10 runs each).

Metric	Legit. eros.	SD	Cop eros.	SD	Cohen's <i>d</i>	<i>p</i> (Welch)
Mean episodes	8.90	1.37	6.90	0.57	+1.91	0.001
Mean active agents	115.0	9.26	147.9	6.18	−4.18	<0.001
Peak active agents	423.9	50.41	461.3	7.45	−1.04	0.044
Share in rebellion	0.767	0.04	0.774	0.02	−0.23	0.619
Mean jailed agents	643.8	20.11	279.7	11.32	+22.32	<0.001
Final cops	64.0	0.00	0.0	0.00		
Final legitimacy	0.003	0.00	0.82	0.00		

Figure 3

Q7 Tocqueville test. Left: legitimacy erosion active agents (red) rise as legitimacy (green) declines. Right: cop erosion rebellion is suppressed for ~150 ticks while cop numbers (purple) remain moderately high, then erupts as enforcement drops below a critical threshold.



Description of results. **Legitimacy erosion** (Figure 3, left) produces progressively intensifying rebellion. As grievance $G_i = H_i(1 - L)$ grows with declining L , more agents cross their activation threshold each tick. Rebellion unfolds as frequent, moderate episodes (8.9 episodes, SD = 1.37) with a mean of 115.0 active agents (SD = 9.26) and a peak of 423.9 (SD = 50.41). Jailed agents average 643.8 (SD = 20.11), because the full complement of 64 cops continues arresting rebels throughout.

Cop erosion (Figure 3, right) produces a markedly different, more Tocqueville-like pattern. For the first ~150 ticks, while cop numbers remain moderately high, rebellion is almost entirely suppressed; agents tolerate sustained oppression. Then, as the number of cops drops below a critical threshold, rebellion erupts suddenly and violently. Despite fewer total episodes (6.9, SD = 0.57), the mean number of active agents is significantly higher at 147.9 (SD = 6.18; $d = -4.18$, $p < 0.001$). The peak reaches 461.3 (SD = 7.45). Jailed agents average only 279.7 (SD = 11.32), because fewer cops remain to make arrests, a massive and significant difference ($d = +22.32$, $p < 0.001$).

The two conditions produce nearly identical overall rebellion shares (76.7% vs. 77.4%; $d = -0.23$, $p = 0.619$), but achieve this through qualitatively different dynamics: legitimacy erosion generates many moderate episodes, while cop erosion generates fewer but more explosive ones.

Interpretation. The key mechanism behind the Tocqueville effect in this model is the estimated arrest probability P_i , defined as the ratio of cops to active rebels within each agent's vision radius. Under legitimacy erosion, grievance increases continuously, but the arrest probability remains high (64 cops are always present), so rebellion builds gradually as individual agents incrementally cross their thresholds. Under cop erosion, grievance stays constant at $H_i(1 - 0.82)$, but the arrest probability drops as cops are removed. Until the cop count falls below a critical threshold, the perceived risk of arrest is sufficient to deter most agents. Once enforcement weakens past this tipping point, the accumulated grievance, unchanged throughout, is suddenly *unleashed*. This creates the paradoxical pattern Tocqueville described: oppression is tolerable as long as it is enforced; the moment enforcement weakens, the population revolts.

The difference in temporal dynamics is informative. Legitimacy erosion produces more episodes ($d = +1.91$, $p = 0.001$) because the escalating grievance crosses individual thresholds at different times, generating waves of rebellion that cops can partially suppress. Cop erosion produces fewer but larger episodes because once cops are depleted, there is no mechanism to suppress rebellion, and it becomes self-sustaining. The dramatically lower jailed count under cop erosion ($d = +22.32$) confirms that the absence of enforcement, not the level of grievance, is the decisive factor.

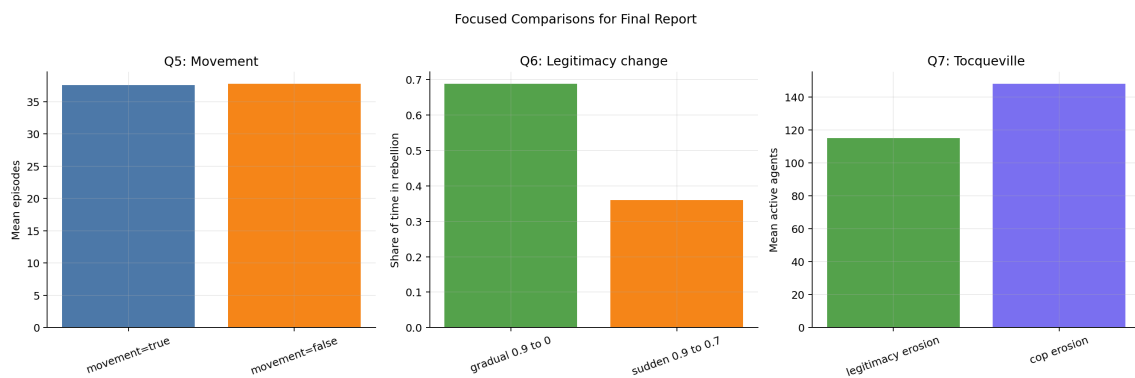
Verification of Tocqueville's claim. Our simulations **support de Tocqueville's observation**. The cop-erosion scenario, where repressive capacity weakens while grievance

remains constant, produces a sudden, explosive outbreak of rebellion that is delayed but ultimately more intense per agent than the legitimacy-erosion scenario ($d = -4.18$ for mean active agents). This directly illustrates de Tocqueville’s paradox: populations endure sustained oppression as long as enforcement holds, but when the government “relaxes its pressure,” they revolt.

This finding is also consistent with the broader simulation literature on threshold-based models of collective action (Epstein, 2002). The model demonstrates that the *source* of regime weakening matters: declining legitimacy and declining enforcement produce qualitatively different rebellion dynamics, even when both lead to similar aggregate rebellion shares ($\sim 77\%$; $d = -0.23$, n.s.). The Tocqueville effect is most visible when enforcement the proximate deterrent weakens, rather than when grievance grows, because the threshold decision in Epstein’s model weights perceived arrest risk heavily. Figure 4 summarizes the key comparisons across Q5–Q7.

Figure 4

Summary comparison across Q5–Q7. Left: movement has negligible effect on episode count. Center: gradual legitimacy erosion produces more total rebellion time than sudden drop. Right: cop erosion produces higher mean active agents than legitimacy erosion.



Question 8**[10 points]**

One difficulty with simulations is validation. Under certain conditions, the outcomes from simulating a model can be validated against reality by looking at historical data. Pick two instances of civic unrest and say if the outcomes of your simulations align with the outcomes of your examples. You will need to familiarize yourself with those instances to a certain degree to be able to compare them against the simulations. Providing your reasoning for why they align with the modeling results or not.

Results

As Sargent (2010) and Gilbert and Troitzsch (2005) emphasize, validation of simulation models against empirical data is essential but challenging. Following Saam (2017)'s distinction between theoretical simulations and empirically validated models, we compare our simulation outcomes with two historical instances of civic unrest in an exploratory fashion.

Case 1: The Tunisian Revolution (2010–2011)

The Jasmine Revolution began on 18 December 2010, when Mohamed Bouazizi's self-immolation in Sidi Bouzid sparked nationwide protests against President Zine al-Abidine Ben Ali's 23-year authoritarian regime. The revolution followed decades of accumulated grievance, high unemployment, corruption, suppression of political freedoms, but was triggered by a sudden catalytic event that dramatically eroded the regime's perceived legitimacy (Campante & Chor, 2012).

Alignment with simulation: This case corresponds closely to our *sudden legitimacy drop* scenario (Q6). The self-immolation and subsequent police violence acted as a sudden shock to perceived government legitimacy, analogous to the instantaneous drop from 0.9 to 0.7 in our simulation. The simulation correctly predicts the observed pattern:

a massive, rapid mobilization (our simulated peak of 631.5 active agents, $SD = 12.7$, mirrors the speed with which protests engulfed the country), followed by either regime collapse or re-equilibration. In Tunisia, Ben Ali fled on 14 January 2011, just 27 days later, suggesting the regime could not survive the initial spike. The model's prediction that the spike is transient (the system stabilizes at the new legitimacy level if the regime survives) also aligns: in other Arab Spring countries where regimes deployed greater force (e.g., Syria, Bahrain), protests were partially suppressed after initial spikes, consistent with the model's equilibration mechanism.

Limitations: The model does not account for international media amplification, the role of social media in coordinating protests, or the military's decision to withdraw support from Ben Ali factors that were decisive in the actual outcome.

Case 2: The East German Peaceful Revolution (1989)

The fall of the Berlin Wall was preceded by Gorbachev's policies of *glasnost* (openness) and *perestroika* (restructuring), which gradually relaxed the repressive grip of the Soviet bloc on its satellite states. East Germans had endured decades of authoritarian rule under the Socialist Unity Party (SED), but beginning in 1988, the relaxation of Soviet military backing signaled that the regime's external enforcement mechanism was weakening. On 9 October 1989, over 70,000 people marched in Leipzig and, crucially, no arrests were made (Pfaff, 1996). This absence of enforcement broke the dam, leading to the opening of the Berlin Wall on 9 November 1989.

Alignment with simulation: This case is a textbook illustration of de Tocqueville's paradox, and aligns remarkably well with our *cop-erosion scenario* (Q7). The gradual withdrawal of Soviet military backing and Gorbachev's explicit signal that force would not be used to suppress protests is analogous to our simulation's gradual reduction of cop numbers. The simulation predicts that rebellion remains suppressed as long as

enforcement capacity exceeds a critical threshold, then erupts suddenly when it drops below. The October 9 Leipzig march, the first large demonstration not met with arrests, corresponds precisely to this tipping point. Our simulation’s finding that cop erosion produces significantly higher mean active agents than legitimacy erosion ($d = -4.18$, $p < 0.001$) also aligns with the massive scale of the 1989 protests once enforcement collapsed.

Limitations: The model treats enforcement as a continuous linear decline, whereas in reality the withdrawal of Soviet support was a discrete political signal. The model also omits emigration pressure (the mass exodus through Hungary), organizational networks (the churches that coordinated protests), and the role of West German media in providing information to East German citizens.

General caveats. As Larooij and Törnberg (2026) note in their discussion of generative social simulation, validation remains the central challenge for agent-based models. Our historical comparisons are exploratory and illustrative, not confirmatory. Both the Epstein model and real-world civic unrest involve far more complexity than any single ABM can capture. Nevertheless, the qualitative alignment between simulation outputs and historical patterns provides suggestive evidence that the model captures important macro-level dynamics of civic unrest, specifically, the distinction between *worsening conditions* and *relaxing repression* as drivers of revolution (Squazzoni et al., 2014).

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